ST06

Genetics: Inheritance and Variation

Expect 2-4 questions from genetics in the Science block. The three certainties are chromosomal disorders (Down, Turner, Klinefelter), ABO and Rh blood groups, and the Mendelian ratios. Sex-linked disorders - colour blindness and haemophilia - are the fourth regular. Everything else in the chapter is insurance.

Genetics looks frightening because of the crosses. It is not. RPSC almost never asks you to work out a cross - it asks you to recall a ratio, a chromosome number, a disorder name or a scientist. Down syndrome as Trisomy 21 has come up word for word in two different papers. ABO blood groups have come up three times. So this chapter is memory work built on a little understanding, and the understanding takes one sitting.

Genetics

Mendel

- Pisum sativum, 7 contrasting pairs, 1866
- Rediscovered 1900 - de Vries, Correns, Tschermak
- Dominance, Segregation, Independent Assortment
- Bateson coined 'genetics', Johannsen 'gene'

Ratios

- Monohybrid F2 - 3:1 pheno, 1:2:1 geno
- Dihybrid F2 - 9:3:3:1
- Test cross - 1:1 and 1:1:1:1
- Incomplete dominance - 1:2:1 both ways

Blood groups

- 3 alleles IA IB i, chromosome 9
- 6 genotypes, 4 phenotypes
- AB codominant, O universal donor
- Rh -ve mother + Rh +ve foetus = erythroblastosis

Sex and sex-linked

- XX-XY human, XX-XO grasshopper
- ZZ-ZW birds - female heterogametic
- SRY on short arm of Y
- Colour blindness, haemophilia A and B

Chromosomal disorders

- Down = Trisomy 21, karyotype 47
- Edwards 18, Patau 13
- Klinefelter 47 XXY, Turner 45 XO
- Barr bodies = X minus 1

DNA

- Watson and Crick 1953, Franklin's X-ray data
- B-DNA right handed, 10 bp/turn, 3.4 nm pitch
- Chargaff A=T, G=C
- Replication semi-conservative, 5' to 3'

Code and mutation

- 64 codons, 61 sense, 3 stop
- AUG start = methionine
- Point vs frameshift
- de Vries mutation theory, Muller X-rays

Applications

- DNA fingerprinting - Alec Jeffreys 1984
- Lalji Singh, CCMB Hyderabad
- HGP 1990-2003, ~20,000-25,000 genes
- Genome India - 10,000 genomes

Mendel - why a monk with peas got it right

Before Mendel, people believed inheritance was a blending of the parents - mix black and white, get grey. Mendel showed that inheritance is particulate. Traits are carried by discrete units that stay intact, separate cleanly and reappear unchanged in later generations. He got there because of three choices. He chose the garden pea, which has clear either-or characters and can self-pollinate. He tracked one character at a time. And he counted - he applied statistics to biology, which nobody had done. He published in 1866, was ignored, and was rediscovered in 1900.

- Mendel worked on the garden pea, Pisum sativum, and took seven pairs of contrasting characters.
- His paper was published in 1866 and rediscovered in 1900, independently, by Hugo de Vries, Carl Correns and Erich von Tschermak.
- The term 'genetics' was coined by William Bateson; the term 'gene' by Wilhelm Johannsen.
- Mendel is called the Father of Genetics.
- Pea suited him because it is naturally self-pollinating, short-lived, and its characters exist in two sharply distinct forms.

The seven contrasting character pairs Mendel used

Character	Dominant form	Recessive form
Stem height	Tall	Dwarf
Flower colour	Violet	White
Flower position	Axial	Terminal
Pod shape	Inflated	Constricted
Pod colour	Green	Yellow
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green

Mendel's three laws - understand them before you state them

Take the three laws in the order the crosses reveal them. The Law of Dominance comes from the F1 generation - cross a tall pea with a dwarf pea and every offspring is tall, so one factor is masking the other. The Law of Segregation comes from the F2 - the dwarf reappears, which means the dwarf factor was there all along, sitting unchanged inside the tall F1 plant, and the two factors of a pair separated cleanly when gametes were formed. Each gamete got one and only one. That is why this law is also called the law of purity of gametes. The Law of Independent Assortment comes only from the dihybrid cross - when you track two characters at once, the way one pair assorts has no effect on the other.

- Law of Dominance - in a heterozygote, one allele expresses itself and masks the other.
- Law of Segregation, also called the law of purity of gametes - the two alleles of a pair separate during gamete formation, so a gamete carries only one.
- Segregation is the only one of the three laws that is universally valid, without exception.
- Law of Independent Assortment - alleles of different gene pairs assort independently of each other.
- Dominance and independent assortment both have well-known exceptions; segregation has none.

TRAP

Independent assortment holds only for genes on different chromosomes, or far apart on the same one. Genes lying close together on the same chromosome are linked and travel together - which is exactly what T. H. Morgan demonstrated in Drosophila. Do not say Mendel's laws have no exceptions.

Monohybrid and dihybrid crosses - the ratios that get asked

A monohybrid cross follows one character; a dihybrid cross follows two. Both start with pure-breeding parents, produce a uniform F1, and reveal the pattern in the F2. What the paper wants is the F2 ratio - phenotypic and genotypic - and it wants the test-cross ratio too. Learn all of them as a block.

How a monohybrid cross runs - tall x dwarf

P generation: TT (tall) x tt (dwarf) → F1: all Tt - every plant tall, the dwarf trait hidden → F1 self-pollinated → F2: 3 tall : 1 dwarf by appearance; 1 TT : 2 Tt : 1 tt by genotype

Every ratio in this section, in one place

Cross	Ratio	Type
Monohybrid F2	3 : 1	Phenotypic
Monohybrid F2	1 : 2 : 1	Genotypic
Dihybrid F2	9 : 3 : 3 : 1	Phenotypic
Monohybrid test cross	1 : 1	Phenotypic and genotypic
Dihybrid test cross	1 : 1 : 1 : 1	Phenotypic and genotypic
Incomplete dominance F2	1 : 2 : 1	Phenotypic, which here equals the genotypic

- A test cross means crossing the unknown individual with the homozygous recessive - it is how you find out whether a dominant-looking plant is TT or Tt.
- A back cross is a cross with either parent; a test cross is specifically with the recessive parent.
- In the dihybrid 9:3:3:1, the 9 shows both dominant traits, the 1 shows both recessive, and the two 3s are the new recombinant combinations.
- Phenotype is what you see; genotype is the allele pair actually carried.

TRAP

3:1 is the F2 phenotypic ratio, 1:2:1 is the F2 genotypic ratio of the same cross - they are two views of one result, not two different crosses. The paper deliberately asks for one and offers the other.

Beyond Mendel - incomplete dominance and codominance

Mendel's dominance is not the only possibility. Sometimes neither allele fully masks the other and the heterozygote comes out in between - red flower crossed with white gives pink. That is incomplete dominance, and the pink is a blend that does not exist in either parent. Sometimes both alleles express themselves fully and separately, side by side, in the same individual - that is codominance, and the AB blood group is the standard example, because an AB person makes both A antigen and B antigen, not some intermediate antigen. Blend versus both-together is the whole distinction.

Incomplete dominance vs Codominance

Incomplete dominance	Codominance
Heterozygote is intermediate between the two parents	Heterozygote shows both parental traits fully and separately
Neither allele is fully expressed	Both alleles are fully expressed
Example - Mirabilis jalapa (four o'clock plant) and Antirrhinum (snapdragon): red x white gives pink	Example - AB blood group in humans, roan coat in cattle, MN blood group
F2 phenotypic ratio 1:2:1, identical to the genotypic ratio	F2 phenotypic ratio also 1:2:1, for the same reason
The new phenotype is a blend not seen in either parent	No new blended phenotype - both parental products appear

- Pleiotropy – one gene producing many effects. Examples: phenylketonuria and sickle cell anaemia.
- Polygenic inheritance – many genes producing one continuously varying trait. Examples: human skin colour and height. Wheat kernel colour was studied by Kolreuter.
- Linkage and crossing over were established by T. H. Morgan working on Drosophila melanogaster.
- Alfred Sturtevant prepared the first genetic map, using recombination frequency as the measure of distance.

YAAD RAKHO

Incomplete dominance = a mixture, like paint – red and white give pink. Codominance = both colours visible side by side, like a checked shirt – A and B antigens both present. Whenever the paper says AB blood group, the answer is codominance, not incomplete dominance.

ABO blood groups – the most reliably asked topic in this chapter

The ABO system is the textbook case of multiple alleles – one gene with more than two alternative forms in the population, though any one person still carries only two of them. There are three alleles: I(A), I(B) and i. I(A) and I(B) are codominant with each other and both are dominant over i. Three alleles taken two at a time give six genotypes, and because of the dominance relations those six collapse into only four phenotypes. Learn the grid, then learn the transfusion rule, then learn Rh – the paper has come at this topic from all three sides.

Six genotypes, four phenotypes

Blood group	Possible genotypes	Antigen on RBC	Antibody in plasma
A	I(A)I(A) or I(A)i	A	anti-B

Blood group	Possible genotypes	Antigen on RBC	Antibody in plasma
B	I(B)I(B) or I(B)i	B	anti-A
AB	I(A)I(B)	A and B both	None
O	ii	None	anti-A and anti-B

- The ABO gene lies on chromosome 9.
- AB is the universal recipient (no antibodies in plasma); O is the universal donor (no antigens on the red cells).
- An AB father and an O mother can only have A or B children - never AB and never O. This is the classic cross question.
- Rh incompatibility - an Rh-negative mother carrying an Rh-positive foetus can make anti-Rh antibodies that attack the foetal red cells.
- The result is erythroblastosis foetalis, also called haemolytic disease of the newborn.
- The first child usually escapes because sensitisation happens at delivery; later Rh-positive pregnancies are at risk. Prevention is anti-D immunoglobulin given to the mother.

ASKED IN RAS

Asked in RAS Pre 2023 - the AB blood group is the best-known human example of codominance, and a statement set on ABO inheritance (three alleles, six genotypes, four phenotypes, I(A) and I(B) codominant) was set the same year. RAS Pre 2021 asked the cross itself: an AB man married to an O woman can have children of group A or B only.

ASKED IN RAS

Asked in RAS Pre 2023 - erythroblastosis foetalis arises when an Rh-negative mother carries an Rh-positive foetus. Blood groups and Rh incompatibility have surfaced in 2013, 2021 and 2023, so treat this as a near-certain area.

Sex determination and sex-linked inheritance

Sex determination is about which parent's gamete carries the deciding chromosome. In humans the mother can only give an X, while the father gives either an X or a Y - so the father's sperm decides the sex of the child. The male here is the heterogametic sex, meaning he makes two kinds of gametes. Birds run the opposite arrangement, where the female is heterogametic, and that reversal is the fact the paper tests. Maleness itself comes from one gene: SRY, on the short arm of the Y chromosome.

Sex determination systems - which sex is heterogametic

System	Found in	Heterogametic sex
XX - XY	Humans, Drosophila	Male
XX - XO	Grasshoppers, some insects	Male
ZZ - ZW	Birds, butterflies	FEMALE
Haplodiploidy	Honeybees - drones haploid (16), females diploid (32)	No sex chromosomes; sex depends on ploidy

- Maleness in humans is determined by the SRY gene on the short arm of the Y chromosome.
- The sperm decides the child's sex - the mother contributes only an X.
- Sex-linked genes lie on the sex chromosomes; almost all the disorders you need are X-linked recessive, because the X carries far more genes than the Y.
- A man has only one X, so a single recessive allele on it expresses itself - which is why X-linked recessive disorders are far commoner in men.

- X-linked recessive disorders - red-green colour blindness, haemophilia A, haemophilia B, Duchenne muscular dystrophy, G6PD deficiency.

- Red-green colour blindness affects about 8% of men and about 0.4% of women.
- Haemophilia A is a deficiency of clotting factor VIII; haemophilia B, also called Christmas disease, is a deficiency of factor IX. Queen Victoria was the famous carrier.
- A colour-blind daughter is possible only if the father is colour blind AND the mother is at least a carrier.
- Criss-cross inheritance - the trait passes from an affected father through his carrier daughter to his grandson.

ASKED IN RAS

Asked in RAS Pre 2021 - a statement set on red-green colour blindness confirming that it is X-linked recessive and commoner in males, and a cross question: a carrier woman married to a normal man has one-half of her sons affected by haemophilia. Work that cross once by hand and it will never leave you.

YAAD RAKHO

Sons get their X only from the mother. So for any X-linked recessive disease, look at the mother to explain an affected son, and look at the father to explain an affected daughter. Birds are the reversal - ZZ male, ZW female, so in birds it is the mother who decides.

Chromosomal disorders - Down syndrome gets the full treatment

A normal human body cell has 46 chromosomes - 22 pairs of autosomes plus one pair of sex chromosomes. Arranged by size and centromere position, that display is a karyotype, and the standard arrangement is the Denver system with groups A to G. Chromosomal disorders arise when a chromosome is present in the wrong number, usually because a pair failed to separate during cell division. That failure is called non-disjunction, and it is the single mechanism behind Down, Turner and Klinefelter alike. Down syndrome is trisomy of chromosome 21 - three copies instead of two, so the total count becomes 47 and not 46. RPSC asked this in identical form in 2018 and again in 2021, which tells you what to do with it: learn the chromosome number, the total count, the person it is named after, and the maternal-age link.

- Down syndrome = trisomy of chromosome 21; karyotype 47, and 47, XX or 47, XY depending on sex.
- First described by Langdon Down; the risk rises sharply with the mother's age.
- Features - short stature, small round head, furrowed protruding tongue, a single palm crease, congenital heart defects, mental retardation.
- It is the commonest chromosomal disorder compatible with survival, and it arises from non-disjunction, not from inheritance of a faulty gene.

Other chromosome-number and structure disorders

Disorder	Chromosome defect
Edwards syndrome	Trisomy 18
Patau syndrome	Trisomy 13
Cri-du-chat syndrome	Deletion of the short arm of chromosome 5
Chronic myeloid leukaemia	Philadelphia chromosome, translocation t(9;22)
Sickle cell anaemia	Not chromosomal - autosomal recessive point mutation

The two sex-chromosome disorders - compare them line by line

Feature	Klinefelter syndrome	Turner syndrome
Karyotype	47, XXY	45, X0
Chromosome count	47	45

Feature	Klinefelter syndrome	Turner syndrome
Sex of the individual	Male, sterile	Female, sterile
Chief features	Tall, gynaecomastia, poor secondary sexual development	Short stature, webbed neck, rudimentary ovaries
Barr bodies	1	0
Cause	Non-disjunction	Non-disjunction

- Number of Barr bodies = number of X chromosomes minus one. Normal female 1, normal male 0, Klinefelter 1, Turner 0.
- Sickle cell anaemia is autosomal recessive - glutamic acid is replaced by valine at the sixth position of the beta-globin chain, a missense point mutation.
- Thalassaemia is a quantitative defect - the globin chain is normal in sequence but made in reduced amounts. Sickle cell is qualitative, thalassaemia is quantitative.

ASKED IN RAS

Asked identically in RAS Pre 2018 and RAS Pre 2021 - Down syndrome is caused by trisomy of chromosome 21, and the affected person has 47 chromosomes. The same papers carried the statement version: described by Langdon Down, incidence rising with maternal age. A fact asked twice in the same form is a fact that will come again.

TRAP

Klinefelter is 47 XXY, a sterile MALE. Turner is 45 X0, a sterile FEMALE. Students reverse them under pressure. Anchor it with the Barr body count - Klinefelter has an extra X, so it has 1 Barr body like a normal female; Turner has only one X, so 0, like a normal male.

DNA - structure, replication and the central dogma

DNA had to be proved to be the genetic material before its structure could matter, and that proof came in stages over twenty-five years. Once Watson and Crick had the double helix in 1953, everything else followed - because the structure itself explains how the molecule copies itself. Two strands, each the exact complement of the other: separate them and each one is a template for rebuilding its partner. Learn the numbers of the helix; the paper asks the numbers.

- The double helix was described by Watson and Crick in 1953, using Rosalind Franklin's and Maurice Wilkins' X-ray diffraction data. Nobel Prize 1962.
- B-DNA is right handed, with 10 base pairs per turn, a pitch of 3.4 nm and a rise of 0.34 nm per base pair.
- Chargaff's rule - A equals T and G equals C. A-T is joined by two hydrogen bonds, G-C by three, so G-C rich DNA is more stable.
- The two strands are antiparallel - one runs 5' to 3', the other 3' to 5'.

How DNA was proved to be the genetic material

Griffith 1928 - transformation in Pneumococcus → Avery, MacLeod and McCarty 1944 - DNA is the transforming principle →

Hershey and Chase 1952 - confirmed with bacteriophage → Watson and Crick 1953 - the double helix →

Meselson and Stahl 1958 - replication is semi-conservative, shown in E. coli

- Replication is semi-conservative - each daughter molecule keeps one old strand and one new one.
- Synthesis always runs 5' to 3'. The leading strand is made continuously; the lagging strand is made in short Okazaki fragments.
- DNA ligase joins the Okazaki fragments - it is the sealing enzyme.
- Reverse transcription, RNA back to DNA, was discovered by Temin and Baltimore - the exception to the one-way central dogma.

Central dogma - Crick's one-way flow of information

DNA → Replication - DNA makes more DNA → Transcription - DNA makes RNA → Translation - RNA makes protein

- The genetic code has 64 codons - 61 sense codons and 3 stop codons, UAA, UAG and UGA.
- AUG is the start codon and also codes for methionine. Methionine (AUG) and tryptophan (UGG) are the two amino acids with a single codon each.
- The code is degenerate (more than one codon per amino acid), non-overlapping, commaless and nearly universal.
- The code was deciphered by Nirenberg, Har Gobind Khorana and Holley - Nobel Prize 1968. Khorana is the Indian-origin name to remember.
- The lac operon, an inducible operon, was explained by Jacob and Monod.

Mutation, variation and DNA fingerprinting

A mutation is a permanent change in the DNA sequence, and it is the ultimate source of all new variation - without it natural selection would have nothing to select from. Mutations are classified two ways: by what happens to the sequence, and by what happens to the protein. Adding or removing bases in numbers not divisible by three shifts the whole reading frame downstream, which is why frameshift mutations are usually far more damaging than a single substitution.

- By sequence change - point or substitution mutation (transition and transversion), and frameshift mutation (insertion or deletion).
- By effect on the protein - silent (no change in amino acid), missense (one amino acid changed, as in sickle cell anaemia) and nonsense (a premature stop codon).
- Physical mutagens - UV light, X-rays, gamma rays. Chemical mutagens - EMS, nitrous acid, 5-bromouracil, acridine dyes.
- Hugo de Vries proposed the mutation theory from Oenothera lamarckiana and used the term 'saltation' for large sudden changes.
- H. J. Muller showed that X-rays induce mutations, working on Drosophila.

- Hardy-Weinberg principle - p + q = 1 and p squared + 2pq + q squared = 1. It says allele frequencies stay constant in an ideal population.
- The equilibrium needs a large, randomly mating population with no mutation, no migration, no genetic drift and no selection. Any real population violates at least one.
- Founder effect and bottleneck are forms of genetic drift.
- Darwin published 'On the Origin of Species' in 1859, after the HMS Beagle voyage and his study of the Galapagos finches - the classic case of adaptive radiation.
- Alfred Russel Wallace reached the same idea independently; Lamarck had earlier proposed inheritance of acquired characters.
- Industrial melanism in the peppered moth Biston betularia is the standard example of directional natural selection.
- Homologous organs indicate divergent evolution; analogous organs indicate convergent evolution.

- DNA fingerprinting was developed by Alec Jeffreys in 1984 and rests on VNTR and STR repeat sequences, which vary enormously between individuals.
- Lalji Singh of CCMB Hyderabad is called the Father of DNA fingerprinting in India.
- Uses - criminal investigation, paternity disputes, identification of disaster victims, wildlife forensics.
- The Human Genome Project ran from 1990 to 2003 and found roughly 20,000-25,000 genes in about 3.2 billion base pairs.

CURRENT LINK

Current link (as of September 2026): the GenomeIndia Project, funded by the Department of Biotechnology, completed whole-genome sequencing of 10,000 Indians, and the dataset was released in January 2025 with the data held at the Indian Biological Data Centre. The point for the exam is the number - 10,000 genomes - and the purpose: India's population carries genetic variation that global reference genomes miss, which matters for disease risk and for drug response.

REVISION RECAP

1. Mendel - Pisum sativum, seven contrasting pairs, published 1866, rediscovered 1900 by de Vries, Correns and von Tschermak.
2. Bateson coined 'genetics', Johannsen coined 'gene'; Mendel is the Father of Genetics.
3. Three laws - Dominance, Segregation (purity of gametes, the only universally valid one), Independent Assortment.
4. Monohybrid F2 - 3:1 phenotypic and 1:2:1 genotypic. Dihybrid F2 - 9:3:3:1. Test cross - 1:1 and 1:1:1:1.
5. Incomplete dominance - Mirabilis jalapa and Antirrhinum, F2 ratio 1:2:1. Codominance - AB blood group, roan cattle, MN group.
6. ABO - three alleles I(A), I(B), i on chromosome 9; six genotypes, four phenotypes; AB universal recipient, O universal donor.
7. Rh-negative mother with an Rh-positive foetus gives erythroblastosis foetalis; prevented by anti-D immunoglobulin.

8. XX-XY human and Drosophila, XX-XO grasshopper - male heterogametic. ZZ-ZW birds and butterflies - female heterogametic. Honeybee drones are haploid.
9. SRY on the short arm of the Y makes a male; the sperm decides the child's sex.
10. Colour blindness affects about 8% of men and 0.4% of women; haemophilia A is factor VIII, haemophilia B (Christmas disease) is factor IX.
11. Down syndrome - trisomy 21, total 47 chromosomes, described by Langdon Down, risk rises with maternal age.
12. Edwards trisomy 18, Patau trisomy 13, cri-du-chat deletion on chromosome 5, Philadelphia chromosome t(9;22) in CML.
13. Klinefelter 47 XXY sterile male with gynaecomastia; Turner 45 XO sterile female with webbed neck; both from non-disjunction.
14. Barr bodies = number of X chromosomes minus one - normal female 1, normal male 0, Klinefelter 1, Turner 0.
15. Watson and Crick 1953 from Franklin's X-ray data; B-DNA right handed, 10 bp per turn, pitch 3.4 nm; Chargaff A=T and G=C.
16. Griffith 1928, Avery-MacLeod-McCarty 1944, Hershey-Chase 1952, Meselson-Stahl 1958 for semi-conservative replication.
17. Central dogma DNA to RNA to protein; reverse transcription found by Temin and Baltimore; ligase joins Okazaki fragments.
18. Genetic code - 64 codons, 61 sense and 3 stop (UAA, UAG, UGA); AUG starts and codes methionine; deciphered by Nirenberg, Khorana and Holley.
19. Point and frameshift mutations; de Vries gave the mutation theory from Oenothera; Muller showed X-rays induce mutation.
20. DNA fingerprinting - Alec Jeffreys 1984, VNTR/STR based; Lalji Singh of CCMB Hyderabad is its Indian father; HGP 1990-2003, about 20,000-25,000 genes.